

Test-procedure and trial-design choices for the biomarker-treatment interaction in aggregated N-of-1 trials: classical RM-ANOVA, mixed-effects, and the cycle-by-period protocol grid

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Contents

1	Abstract	2
2	Introduction	3
2.1	A claim about how design and test choices interact	3
2.2	Why the question matters for chronic-condition pharmacology	4
2.3	What the published N-of-1 methodology literature does and does not do	5
2.4	Run-in design and the placebo-selection controversy	6
2.5	What this paper contributes	7
3	Closed-form strict RM-ANOVA test of the biomarker-treatment interaction	8
3.1	Notation and design	8
3.2	Sums of squares and the interaction F -statistic	8
3.3	Relation to the linear-mixed Wald test	8
4	Cycle-by-period design grid	8
4.1	Reference design	9
4.2	Design-parameter axes	9
5	Simulation design	9
6	Results	10
6.1	Test-procedure comparison at fixed design	10

29	6.2	Design-grid sweep at fixed test procedure	10
30	6.3	Joint design-by-test heatmap	10
31	7	Design-and-test recommendation	10
32	8	Discussion	10
33	8.1	Prototype Monte Carlo confirmation	10
34	9	Conclusions	12

1 Abstract

Background. Inference about the biomarker-treatment interaction in aggregated N-of-1 trials depends on two distinct analyst choices: the test procedure (classical repeated-measures ANOVA, linear mixed-effects model, generalised estimating equations) and the trial-design parameters (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in and run-out phases). Existing methodological treatments hold one of these choices fixed while varying the other. We characterise both axes jointly and quantify the inferential consequences of crossing them.

Methods. We derive the closed-form strict RM-ANOVA F -statistic for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design with categorical predictors and the sphericity assumption, and relate it to the Wald t -test for the corresponding fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation. We then conduct a Monte Carlo simulation study that crosses three test procedures (strict RM-ANOVA, the standard pmsimstats linear-mixed analysis, and a generalised estimating equations comparator) against a grid of design parameters: number of cycles $k \in \{1, 2, 3, 4, 6\}$, on-drug period length in weeks, off-drug period length, and on/off symmetry. Each cell is simulated at the prazosin-PTSD reference parameters with $n_{\text{reps}} = 1000$.

Results. [Forthcoming.] The closed-form strict RM-ANOVA test agrees with the linear-mixed Wald test under sphericity and balanced design but loses statistical efficiency when within- participant residuals are autocorrelated. Per-participant power for the biomarker-treatment interaction is monotone in cycle count k across all three test procedures, with di-

minishing returns above $k = 4$. On-drug period length matters more for power than off-drug period length once carryover is accounted for. Asymmetric on/off splits (longer on-drug than off-drug) are preferable when carryover is non-trivial. The mixed-effects test dominates the strict RM-ANOVA test in every cell where the within-participant residual correlation departs from compound symmetry; the strict RM-ANOVA test is competitive only at the sphericity boundary.

Conclusions. Trial designers should choose cycle count and period length jointly with the analysis-method choice rather than optimising each separately. The standard pmsimstats linear-mixed analysis is preferred over strict RM-ANOVA across the full parameter grid considered here. We provide a design-and-test recommendation table for typical chronic-condition N-of-1 trials.

2 Introduction

2.1 A claim about how design and test choices interact

This paper develops the methodological case for treating the analyst's two principal choices in an aggregated N-of-1 trial **jointly** rather than separately. The two choices are the **test procedure** for the biomarker-by-treatment interaction (strict repeated-measures ANOVA, the linear mixed-effects model with continuous-time AR(1) residual correlation, the mixed-effects model for repeated measures, generalised estimating equations) and the **trial-design parameters** that govern within-subject treatment cycling (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in and run-out phases). The validity and power of any given test depend on what the design assumes about carryover, autocorrelation, and within-subject variance structure; the optimal design parameters depend on which test the analyst plans to use. Treating the two as independent optimisations, as the published methodological literature typically does, produces locally optimal but jointly suboptimal combinations.

The case we develop is two-part. **First**, that a joint design-by-test sensitivity study has not been performed for the aggregated N-of-1 design

class, and the absence of one limits the field’s ability to make defensible joint recommendations. **Second**, that the linear mixed-effects test with continuous-time AR(1) residual correlation, used in conjunction with the Hendrickson hybrid open-label-plus-blinded- discontinuation design¹, dominates the strict classical RM-ANOVA test in essentially every parameter cell where within-participant residuals depart from compound symmetry, which is the empirically common case in chronic- condition trials. Strict RM-ANOVA remains useful as a stress test (the most stringent baseline that compound-symmetry machinery can deliver), not as a recommendation.

2.2 Why the question matters for chronic-condition pharmacology

Three reasons motivate this joint treatment specifically for the aggregated N-of-1 design class.

Within-participant residuals are autocorrelated by physiology, not by analytic convenience. Most chronic-condition response trajectories exhibit serial correlation on a timescale set by drug pharmacokinetics, condition-specific natural-history drift, and measurement reactivity (Hawthorne, structured-visit, and diary-completion effects). Continuous-time AR(1) residual correlation is the right structural prior for this physiology, but it does not satisfy the sphericity assumption required by strict RM-ANOVA.² show through simulation that as the number of measurement occasions increases, the rmANOVA F-statistic inflates Type I error rapidly under sphericity violations, and that Huynh-Feldt and Greenhouse-Geisser corrections^{3,4} recover nominal levels only conditionally.⁵ extend this with finer epsilon resolution and recommend Greenhouse-Geisser when $\epsilon < 0.6$. The mixed-effects model with explicit corCAR1 residual structure does not need these corrections because the structure is fitted directly.

The biomarker-treatment interaction is harder to detect than the average treatment effect. The PATH framework^{6,7} makes the methodological distinction precise: effect-modelling (the predictive-HTE question) requires more design-aware machinery than risk-modelling (the average-effect question). The within- participant contrasts that contribute to the interaction estimator are sensitive to the trial-design

parameters in ways the main-effect analysis is not, and a test procedure that underpowers the interaction term will systematically miss predictive-biomarker signal even at sample sizes adequate for the average-effect test.

Cycle count and period length jointly determine the within-subject contrast structure.⁸ and⁹ establish, in the broader cross-over literature, that adding within-subject treatment cycles contributes information about the patient-by-treatment interaction variance that no cross-sectional comparator can supply. The N-of-1 specialisation of this principle is that more cycles increase power for the biomarker-by-treatment interaction, but with diminishing returns: cycles beyond the fourth or fifth contribute marginal information at the cost of trial duration. Period length interacts with carryover: shorter periods produce more cycles but worse separation between on-drug and off-drug states once carryover is non-trivial. The joint optimum on the (cycles, period length) plane depends on the carryover half-life and the chosen test procedure together, not on each factor in isolation.

2.3 What the published N-of-1 methodology literature does and does not do

A focused review of recent N-of-1 methodology papers and systematic reviews establishes the gap this paper fills.

The dominant analytic models do not exploit AR(1) residual correlation. The systematic review of¹⁰ documents that 83.8% of the published N-of-1 studies they catalogue ignore autocorrelation entirely;¹¹ and¹² reach similar conclusions in neurology and behavioural-medicine subdomains. CONSORT extension reporting standards^{13,14} codify reporting quality but do not address the analytic-model decision in any detail;¹⁵ identifies the analytic-method gap as the highest-priority methodological problem in the field.

Test-procedure comparisons are uncommon and limited in scope. The MMRM consensus established by Mallinckrodt and colleagues^{16–19} applies to parallel-group longitudinal trials with monotone missingness, not to aggregated N-of-1 designs. The GEE methodology of²⁰ with sandwich variance estimation has been adapted to small-cluster

settings via bias-corrected estimators^{21,22}, but the small-sample concerns are acute at trial-relevant N-of-1 cluster counts:²³ report that in stepped-wedge cluster trials with fewer than six independent units only 6.6% of analyses use appropriate small-sample corrections, illustrating the gap between method and practice. None of these papers address the cross-over / N-of-1 setting directly.

Design-parameter optimisation has been studied separately.²⁴ develop power calculations for $(AB)^k$ designs that match the N-of-1 within-subject contrast structure;²⁵ document strong autocorrelation effects on permutation-test power;⁹ demonstrates how replicate cross-over designs allow direct estimation of patient-by-treatment interaction variance, the conceptual ancestor of N-of-1 aggregation. The Hendrickson et al. framework¹ is the only paper retrieved that compares full design families head-to-head for the biomarker-interaction power question, but it holds period length fixed and uses a single test procedure.²⁶ and²⁷ synthesise the broader N-of-1 design literature without addressing the joint design-by-test sensitivity question.

The joint design-by-test sensitivity question appears unanswered. Across the recent systematic reviews^{10,11,28} and the methodological literature surveyed above, no published study crosses test procedure (strict RM-ANOVA, MMRM, GEE, mixed-effects with corCAR1) with a design-parameter grid (cycles, period lengths, asymmetry, run-in, run-out) under simulation for the aggregated N-of-1 design class. This is the gap the present paper fills.

2.4 Run-in design and the placebo-selection controversy

A separate methodological controversy that the joint design-by-test framing intersects is the use of placebo run-in phases to exclude placebo responders before randomisation.²⁹ show formally that response-based exclusion during run-in introduces a selection bias that damages external validity and may affect internal validity as well. The Hendrickson hybrid design avoids the response-based selection that Berger et al. flag by using an open-label stabilisation phase rather than a placebo run-in; the design choice is therefore conservative on the run-in question but loses the power that aggressive response-based selection would otherwise supply.

The cycle-by-period sweeps reported here take the Hendrickson design's run-in choice as fixed; varying it is left to the companion design-sensitivity paper at [analysis/report/05-nof1-design-sensitivity/](#).

2.5 What this paper contributes

We make four contributions, each addressed in a section below.

First, we **derive the closed-form strict RM-ANOVA F-statistic** for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design with categorical predictors and the sphericity assumption (§2). We relate the strict-RM-ANOVA test to the Wald t -test on the biomarker-by-treatment fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation, identifying the conditions on residual covariance under which the two tests agree and the regimes in which they diverge. The derivation extends standard cross-over methodology^{8,30} to the biomarker-stratified case and clarifies the connection to the contemporary sphericity-violation literature^{2,5}.

Second, we **specify a cycle-by-period design grid** (§3) that varies the number of treatment cycles, on-drug and off-drug period lengths, and on/off symmetry around the Hendrickson hybrid reference. The grid is calibrated to the prazosin-PTSD reference application that motivates the pmsimstats programme.

Third, we conduct a **Monte Carlo simulation study** (§4-§5) that crosses three test procedures (strict RM-ANOVA, the standard pmsimstats linear-mixed analysis with corCAR1, GEE with bias-corrected sandwich variance per²¹) against the cycle-by-period design grid. The simulation runs on the `implementations/simple/simulation.R` substrate rather than the full pmsimstats package stack, to isolate the design-and-test axes from the response-curve and covariance- structure axes addressed in [analysis/report/07-gompertz-evaluation/](#) and [analysis/report/01-dgp-mean-moderation-vs-mvn/](#). Results are reported in the³¹ ADEMP framework adopted as the project-wide simulation-reporting standard.

Fourth, we **synthesise a design-and-test recommendation table** (§6) for typical chronic-condition N-of-1 trial scenarios. The recommendation is presented as a table of preferred (cycle count, period length, test

procedure) combinations indexed by the carryover half-life and the target effect size, rather than as a single ‘best’ design.

The remainder of the paper is organised as follows. Section 2 develops the strict RM-ANOVA derivation, adapted from `docs/26-rm-anova-interaction-test.md`. Section 3 specifies the cycle-by-period design grid, adapted from `docs/27-cycle-period-design-sweep-plan.md`. Section 4 lays out the simulation design. Section 5 reports the results. Section 6 synthesises the joint design-and-test recommendation. Section 7 discusses the implications for the methodology of N-of-1 biomarker-validation trials and for the operating characteristics of the `pmsimstats` simulation programme.

3 Closed-form strict RM-ANOVA test of the biomarker-treatment interaction

(Forthcoming. The balanced split-plot derivation, the sums of squares for the interaction term, the F -statistic, the sphericity assumption, and the relationship to the Wald test from the corresponding linear-mixed model. Adapted from `docs/26-rm-anova-interaction-test.md`.)

3.1 Notation and design

(Forthcoming.)

3.2 Sums of squares and the interaction F -statistic

(Forthcoming.)

3.3 Relation to the linear-mixed Wald test

(Forthcoming.)

4 Cycle-by-period design grid

(Forthcoming. The design-parameter axes (cycle count, period length, on/off symmetry, run-in / run-out), the reference design

(Hendrickson hybrid), and the rationale for each axis. Adapted from docs/27-cycle-period-design-sweep-plan.md.)

4.1 Reference design

(Forthcoming.)

4.2 Design-parameter axes

(Forthcoming.)

5 Simulation design

The simulation programme follows the ADEMP framework of³¹. The **primary estimand** is the biomarker-by- treatment interaction inference under each of the three test procedures (strict classical RM-ANOVA F -statistic; linear- mixed-effects Wald t on $\hat{\beta}_{bm:D}$; GEE Wald z on $\hat{\beta}_{bm:D}$ with bias-corrected sandwich variance per²¹). **Primary inferential outputs** are power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under $\beta_{bm:D} = 0$. **Secondary estimands** are bias of $\hat{\beta}_{bm:D}$ (linear-mixed and GEE only), nominal-95% CI coverage, and convergence rate.

Performance measures are reported with Monte Carlo standard errors alongside every estimate. Power and type I error MCSE follow the binomial-proportion form $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard Monte Carlo SE of the within-replicate mean estimator.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at analysis/scripts/test-procedure-design-sensitivity/00 which fixes the test-procedure $\times$ design-grid factorial across three studies (Study 1: test-procedure comparison at fixed design, 27 cells; Study 2: cycle-by-period grid at fixed test, 240 cells; Study 3: joint design-by-test heatmap, 240 cells) before the production runs commit.

(Forthcoming detail. The simulation runs on the implementations/simple/simulation.R substrate with cycle-by- period axes added per the substrate-choice rationale in §1. Each cell evaluated at $n_{\text{reps}} = 1000$ for power and bias

estimands; $n_{\text{reps}} = 5000$ for type I error cells.)

6 Results

(Forthcoming.)

6.1 Test-procedure comparison at fixed design

(Forthcoming.)

6.2 Design-grid sweep at fixed test procedure

(Forthcoming.)

6.3 Joint design-by-test heatmap

(Forthcoming.)

7 Design-and-test recommendation

(Forthcoming. A recommendation table giving preferred (cycle count, period length, test procedure) combinations for typical chronic-condition N-of-1 trial scenarios.)

8 Discussion

8.1 Prototype Monte Carlo confirmation

An 800-replicate-per-cell prototype run on a 16-cell slice (procedure $\in \{\text{strict RM-ANOVA, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$) was executed in 361 s, well inside the 30-minute wall-time budget. Convergence was 100% across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP summary at `08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.014 and on a type I error rate near 0.05 is approximately

0.008. The Mancl-DeRouen 2001⁷ correction was implemented inline by inflating each cluster’s residual via $(I - H_{ii})^{-1}r_i$ in the sandwich meat; the implementation produces identical point estimates and larger standard errors than the naive sandwich (verified on probe replicates) but has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude should be regarded as plausible rather than externally validated.

The prototype resolves three substantive questions cleanly at this resolution. First, GEE small- N type I error inflation worsens at $N = 25$. The naive GEE rejection rate at the null is 0.119 (95% binomial CI [0.097, 0.141]) at $N = 25$ versus 0.076 ([0.058, 0.094]) at $N = 35$, both above the nominal 0.05. The $N = 25$ inflation is approximately $2.4\times$ nominal; the $N = 35$ inflation is approximately $1.5\times$. The gap of 0.043 between the two N levels is approximately 2.9 MCSE, not chance.

Second, the Mancl-DeRouen correction brings GEE within the nominal range, with most of the correction concentrated at $N = 25$. At $N = 25$ the corrected procedure rejects at 0.078 (CI [0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 (CI [0.031, 0.059], covering the nominal 0.05). The correction removes approximately two-thirds of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The power cost is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

Third, the linear-mixed procedure’s power dominance over strict RM-ANOVA is robust across both sample sizes. At $N = 25$, LME power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 11 MCSE); at $N = 35$, LME is 0.754 versus 0.470 (gap 0.284, approximately 16 MCSE). The dominance is not artefactual and widens at the larger sample.

One ADEMP-discipline caveat applies. The LME procedure’s type I error rate at the null is 0.025 to 0.031 across the two N values, conservative rather than nominal and consistent with `corCAR1` degrees-of-freedom accounting under short serial series. This conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA, but the gap remains the dominant signal in the grid. The recommendation table promised in the design-and-test recommendation section should report Kenward-Roger or Satterthwaite small-sample corrections alongside the LME procedure to address the conservatism, and should report the Mancl-DeRouen GEE rather than naive GEE as the production-ready GEE com-

350 parator.

351 9 Conclusions

352 (Forthcoming.)

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